Assignment - 1

NLP: CS60075
Autumn Semester 2025
IIT Kharagpur

Enhancing Figurative Language Recognition Using POS Tagging

Sample Code [You may or may not use stuff from this. It's just some boiler-plate code] - https://colab.research.google.com/drive/1sfARw_asCMJwiFIGIUBUernwnuC8uoaN?usp=sharing

Point of contact - Naquee Rizwan (naqueerizwan1998@gmail.com)

Assignment Beginning: 04:00 PM, 12th August 2025

Assignment Deadline: 11:59 PM, 31st August 2025

Plagiarism Alert

- Plagiarism in any form is strictly prohibited.
- Both the SOURCE and the TARGET individuals will suffer a penalty of 100% if a plagiarism detector tool detects 60% or more code similarity.
- You are not supposed to share your assignment with anyone even after the deadline for submission is over.
- We shall use multiple AI detectors like GPT (and 3 more undisclosed), and if at least 3 of them flag your code as AI-generated then we shall deem the submission to be 100% plagiarism.

OVERVIEW

The objective of this assignment is to deepen your understanding of Natural Language Tagging or more specifically Part-of-Speech (POS) tagging which involves assigning grammatical categories (such as nouns, verbs, and adjectives) to words in a sentence.

One of the pivotal NLP tasks is Figurative Language Recognition which involves determining the figurative language expressed in a piece of text or image, especially in order to determine the owner's/viewer's state of mind towards a particular topic, product, etc. Some typical figurative language types incorporate humor, metaphor, sarcasm, simile and idiom.

While POS tagging might seem like a preliminary linguistic task, its impact extends far beyond syntax. By accurately identifying the grammatical roles of words in a sentence, POS tagging contributes to understanding the context, semantics, and even figurative language conveyed by the text. This is where the concept of this assignment emerges – the exploration of the integration of POS tagging and figurative language recognition.

We have seen that some words can be used as multiple parts of speech in the English language. For example:

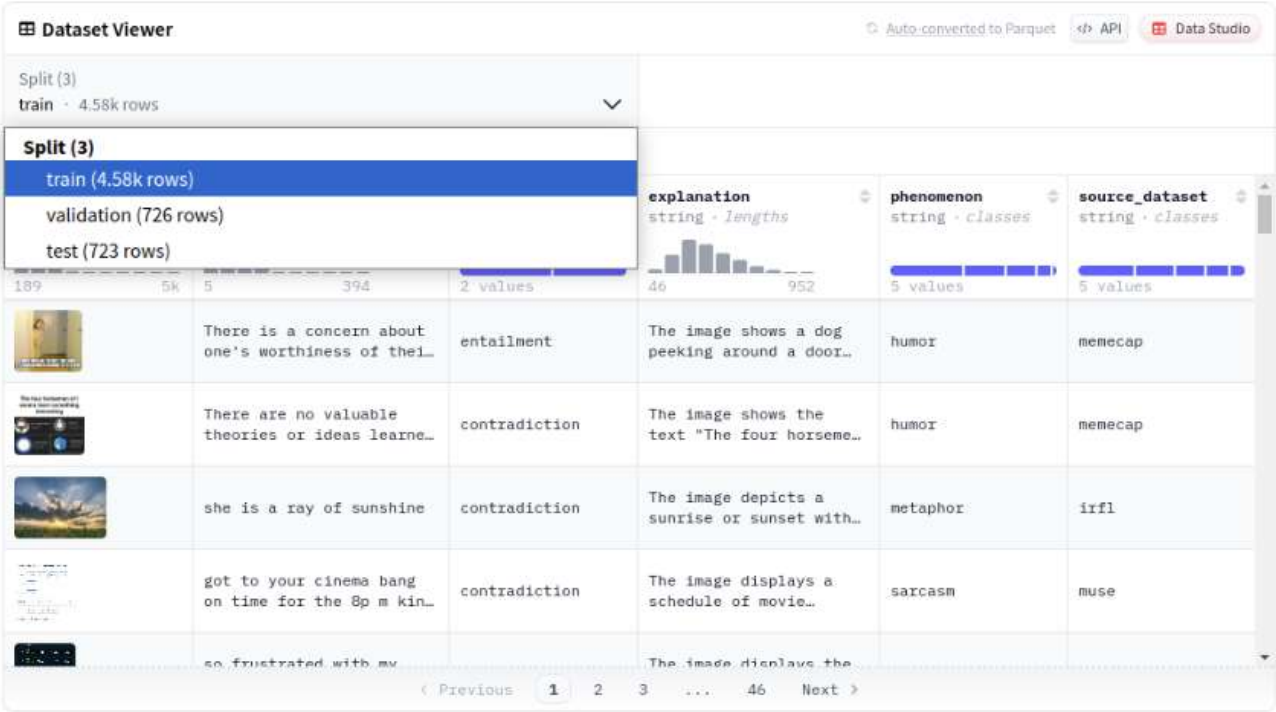
*"Kevin has dark hair and **fair** skin."*	*(ADJECTIVE)*
*"The new **fair** is boring."*	*(NOUN)*

The addition of POS tags introduces a new dimension of linguistic insight that may significantly enhance figurative language recognition accuracy and depth.

You will implement a POS tagging system, apply it to a Classical Figurative Language Recognizer, and evaluate its impact on figurative language recognition accuracy.

TASKS

1. **POS Tagger** Implementation **(from scratch).** **[40% of TOTAL]**
   - Use the treebank corpus (from nltk) for training data. [nltk downloader]
   - Implement the Viterbi Algorithm (dynamic programming) for POS Tagging. **[It is mandatory to implement the algorithm yourself. No in-built library should be used for this task.]**
   - **Keep all the POS tags for this task as that will give you proper transition and emission probabilities.**
2. **Vanilla Figurative Language Recognizer** **[25% of TOTAL]**
   - Use the V-FLUTE corpus for data. [Use Hugging Face APIs to get the dataset. In the shared dataset's link, train/validation/test splits are already present. Strictly use only "split" Subsets (as per the link shared above) for training, validation and testing.] [**Note:** You will have to accept the license to use the dataset and will have to login through access token (use write access to avoid any intermittent errors) to make it work in your code snippets.]



- Regarding the columns, don't worry about the images and other columns. Only take the **"claim" and "explanation" columns** as input (be creative in using these two input columns effectively) and the **"phenomenon" column** as your required output for training, validation and testing.
- Use Tfidf for creating sentence embeddings.
- Train a classical Classifier (Naive Bayes or SVM from sklearn) for figurative language recognition using the above features. Each text has to be classified into one amongst the five labels (humor, metaphor, sarcasm, simile or idiom). Kindly look into the dataset for more details.

3. **Improved Figurative Language Recognizer** **[30% of TOTAL]**
   - Use the POS Tagger in Task 1 for POS tagging the dataset.
   - Implement a pipeline to integrate the POS tag features along with the sentence embeddings. (Use your own creativity and **mention your strategy in the report**)
   - Train the same Classifier (as chosen in Task 2) again for figurative language recognition using the new features.

4. **Report** **[5% of TOTAL]**
   - Add your observations to a report (submit in pdf format).
   - Compare the performance of your POS-tag-enhanced model with a baseline model that doesn't use POS tags. **[Don't worry about scores]**
   - Make sure to include the classification reports and confusion_matrix of both models in the report for the test split.

- Make sure to highlight any advanced modifications that you've done in your report.

Learning Outcomes

- POS Tagging
- Figurative Language Recognition
- Custom Pipeline with Additional Features
- Work Presentation

Submission Guidelines

- You have to use IPython Notebooks for coding. (Use Google Colab, Kaggle for running your assignments).
- Include small documentation on each cell of the notebook. Further, properly add comments to your code as and when required.
- **Important –** Make sure your IPython Notebook has the outputs from each cell. The classification report and confusion matrix must be present in the submitted notebook as well. Absence of these results will lead to straight **deduction of 80% marks.**
- You have to submit the IPython notebook and Assignment report as mentioned above. Name each of them as NLP_Assignment_1_Roll_No.ipynb and NLP_Assignment_1_Roll_No.pdf, respectively.

  [**For eg:** NLP_Assignment_1_23CS91R06.ipynb, NLP_Assignment_1_23CS91R06.pdf]

- Submit these two files in **.zip format** on **Google Form**. **Note:** Your zip file must be named NLP_Assignment_1_Roll_No.zip and if you submit more than once, only your **latest submission** will be considered. **Note:** Submit in **.zip** only. Any other format (like **.tar**) will have your assignment being unchecked. Straight away **0 marks will be awarded** in this case.

  [**For eg:** NLP_Assignment_1_23CS91R06.zip]

HAPPY LEARNING